

Joint Gravity and Radiometric Geophysical Inversion via Quantum Algorithms



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Abstract

This report presents a staged study on joint gravity and radiometric forward modeling for geophysical inversion, with emphasis on a hybrid quantum-classical computational framework. The central objective is to estimate an interpretable subsurface model from gravity and radiometric observations by combining physically motivated forward modeling, data calibration, stochastic uncertainty analysis, and quantum-parameterized regression or inversion components. The study is organized into four stages. Stage 0 develops quantum linear and cubic regression models on controlled synthetic datasets in order to establish the regression and feature-map methodology. Stage 1 constructs a synthetic geophysical model containing host rock, a dense body, a faulted basin, and a shallow soil layer, and validates the inversion procedure against known model parameters. Stage 2 applies the calibrated workflow to a selected 10 km field-data window. Stage 3 extends the interpretation to the complete 30 km profile by performing three independent 10 km inversions and stitching the resulting geological sections.

The synthetic inversion in Stage 1 recovered the principal model parameters with small relative errors: approximately 6.2% for the fault position, 3.4% for dense-body density contrast, 3.2% for basin-density contrast, and 4.0% for dense-body top depth. For the calibrated real-data inversion in Stage 2, the selected 0–10 km window gave a gravity fit of $R^2 = 0.989$ and $\text{RMSE} = 1.267$ mGal. The corresponding radiometric fits for K, U, and Th gave $R^2 = 0.951$, 0.929 , and 0.891 , respectively. In Stage 3, the stitched full-profile interpretation gave a unified gravity fit of $R^2 = 0.965$ and $\text{RMSE} = 1.756$ mGal, while the stitched K, U, and Th fits gave $R^2 = 0.883$, 0.876 , and 0.866 . These results show that a direct synthetic-to-field transfer is insufficient, whereas a calibrated and survey-aware formulation can produce quantitatively strong fits and geologically interpretable models.

1. Introduction

1.1. Background

Geophysical inversion is the process of estimating subsurface physical parameters from observations recorded at or near the Earth’s surface. In gravity surveying, the measured anomaly is produced by lateral and vertical variations in density. A positive density contrast generally increases the gravity response, whereas a low-density sedimentary basin or weathered zone generally reduces it. In radiometric surveying, the observed gamma-ray response is related primarily to the near-surface abundance of potassium (K), uranium (U), and thorium (Th). These two data types therefore provide complementary information. Gravity is sensitive to density variations at greater depth, whereas radiometric observations are much shallower and are strongly affected by surface composition, weath-

ering, and soil cover.

The inversion of geophysical data is inherently non-unique. Many different density distributions can produce similar gravity anomalies, and shallow radiometric highs do not by themselves define deep geological structure. This non-uniqueness is reduced when geological assumptions, parameter bounds, and multiple data types are included in a common objective function. In the present work, gravity and radiometric data are interpreted jointly for a faulted-basin type model. The aim is not only to fit the measured curves, but also to obtain a subsurface section that remains consistent with basic geological expectations.

1.2. Problem Statement

The project addresses the problem of recovering an interpretable two-dimensional subsurface section from gravity and radiometric profiles. The main model consists of a background host rock, a denser body, a low-density basin, and a shallow soil or near-surface layer. In the synthetic stage, the true model is known and is used to test whether the inversion recovers the correct geometry and parameter values. In the real-data stages, the true subsurface model is unknown, so the quality of the inversion is assessed using data fit, physical plausibility, stability under multiple starts, and consistency across adjacent windows.

The real field profile introduces additional complications that are absent in the synthetic problem. The Bouguer gravity curve contains a regional component that must be removed before the residual anomaly can be interpreted locally. The observed anomaly may also have different amplitude and polarity from the synthetic convention. Similarly, real radiometric measurements reflect surface processes and acquisition conditions that are not captured by a simple step-like synthetic model. For this reason, the real-data workflow requires robust trend removal, scale-offset calibration, optional profile reversal, and a radiometric response model that can represent localized shallow enrichment.

1.3. Objectives

The specific objectives of the study are as follows:

1. to implement quantum feature-map regression for linear and cubic functions as a controlled preliminary stage;
2. to construct a physically interpretable synthetic gravity and radiometric forward model for a faulted-basin geometry;
3. to perform quantum-parameterized inversion on the synthetic model and evaluate recovery accuracy against known parameters;

4. to adapt the inversion workflow for real field data through regional-trend removal, survey calibration, and real-style radiometric response modeling;
5. to extend the selected 10 km inversion strategy to a full 30 km profile using three independently inverted 10 km windows;
6. to evaluate the strengths, limitations, and geological meaning of the recovered models.

1.4. Organization of the Work

The work is divided into four computational stages:

1. **Stage 0: Quantum Regression.** Quantum feature-map regression is tested on generated linear and cubic datasets.
2. **Stage 1: Synthetic Forward Modeling and Inversion.** A known geological model is forward modeled and inverted.
3. **Stage 2: Real-Data 10 km Inversion.** The best 10 km field-data window is inverted after calibration.
4. **Stage 3: Full-Profile Stitched Inversion.** Three independent 10 km inversions are combined to interpret the 30 km profile.

2. Geophysical Basis of the Study

2.1. Gravity Anomaly and Density Contrast

Gravity surveying measures variations in gravitational acceleration caused by subsurface density contrasts. If $\Delta\rho(\mathbf{r})$ represents the density contrast relative to a reference background, the vertical gravity anomaly at an observation point $\mathbf{r}_0$ can be expressed in integral form as

$$\Delta g_z(\mathbf{r}_0) = G \int_V \Delta\rho(\mathbf{r}) \frac{z - z_0}{|\mathbf{r} - \mathbf{r}_0|^3} dV, \quad (1)$$

where G is the gravitational constant and V is the volume containing the anomalous mass. For numerical modeling, the subsurface is divided into small rectangular cells or prisms. The total anomaly is calculated by summing the contribution from all cells. This discretized representation is commonly used in applied gravity interpretation because it allows complex shapes to be approximated by a finite set of simple bodies [5, 1].

The synthetic model used here is a 2.5D model. In such a model, the subsurface section varies along the profile distance and depth, while the bodies are assumed to extend a finite distance perpendicular to the profile. This is more realistic than a purely two-dimensional infinite-strike model, but it is still simpler than a fully three-dimensional

inversion. The model response is therefore interpreted as a profile-scale approximation rather than a complete 3D geological reconstruction.

2.2. Regional and Residual Gravity

Field Bouguer gravity data commonly contain both regional and residual components. The regional component is produced by broad, long-wavelength geological structures or crustal-scale variations. The residual component is associated with shorter-wavelength features, such as local basins, contacts, faults, or dense bodies. Since the forward model in this project is local, the inversion is performed on the residual gravity anomaly rather than on the raw Bouguer anomaly.

The residual is obtained by fitting a robust second-degree polynomial trend to the full Bouguer profile and subtracting it from the observed data:

$$g_{\text{res}}(x) = g_{\text{obs}}(x) - g_{\text{reg}}(x). \quad (2)$$

The choice of a low-order polynomial is a compromise. A very low-order trend may leave regional effects inside the residual, while a very high-order trend may remove genuine geological signal. The degree-2 trend used in this work provides a smooth regional field and leaves the local anomaly shape required for inversion.

2.3. Radiometric Response

Airborne or ground radiometric measurements record gamma-ray emissions associated mainly with K, U, and Th. These elements are useful geological indicators because their concentrations vary between lithologies and alteration zones. However, gamma rays are strongly attenuated, so the measured response is dominated by the uppermost part of the subsurface. For that reason, radiometric data should not be interpreted as direct evidence of deep density distribution. In the present workflow, the radiometric channels are used as shallow compositional constraints that help locate contacts and enrichment zones.

The synthetic radiometric model assigns characteristic K, U, and Th values to host rock, basin, and soil units. The real-data model is more flexible. It includes a localized radiometric enrichment term, smooth contact transitions, and simple calibration coefficients. This modification was necessary because the field radiometric curves contain localized peaks that cannot be represented by a purely step-like synthetic response.

2.4. Joint Inversion

The inversion seeks a parameter vector $\mathbf{m}$ that minimizes the mismatch between observed and predicted gravity and radiometric responses. A general normalized objective function

can be written as

$$\Phi(\mathbf{m}) = w_g \frac{1}{N_g} \sum_{i=1}^{N_g} \left[\frac{g_i^{\text{cal}}(\mathbf{m}) - g_i^{\text{obs}}}{\sigma_g + \epsilon} \right]^2 + w_r \sum_{c \in \{K, U, Th\}} \frac{1}{N_c} \sum_{i=1}^{N_c} \left[\frac{r_{c,i}^{\text{cal}}(\mathbf{m}) - r_{c,i}^{\text{obs}}}{\sigma_c + \epsilon} \right]^2. \quad (3)$$

Here g denotes gravity, r_c denotes the radiometric response for channel c , w_g and w_r are relative weights, and σ values normalize the residuals. The calibration superscript indicates that the predicted curve may be scaled and shifted before comparison:

$$d^{\text{cal}} = ad^{\text{pred}} + b. \quad (4)$$

This calibration is particularly important for real data, where the model response and survey response may differ in sign, amplitude, or baseline.

3. Quantum and Computational Framework

3.1. Motivation for Quantum Methods

Quantum algorithms are of interest in inverse problems because many inverse problems involve high-dimensional search, nonlinear objective functions, and repeated forward evaluations. Current quantum computers are still limited by noise and qubit count, so the present work uses a hybrid quantum-classical approach. The quantum component is used either to generate nonlinear features for regression or to parameterize bounded model variables through quantum expectation values. The optimization and data handling remain classical.

The purpose of using quantum methods in this project is therefore methodological. The study demonstrates how quantum feature maps and variational quantum circuits can be integrated into geophysical modeling workflows. It does not claim a computational speedup over classical inversion for the present problem size. Instead, it establishes a reproducible framework that can later be extended to quantum kernels, hardware execution, or larger hybrid models.

3.2. Quantum Feature-Map Regression

In the regression stage, a scalar input x is encoded into a quantum state $|\psi(x)\rangle$ by applying parameterized rotations and entangling gates. Measurements of Pauli-Z expectation values are then used as nonlinear features. A typical feature vector has the form

$$\phi(x) = [1, x, x^2, \langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_0 Z_1 \rangle, \dots]. \quad (5)$$

The predicted output is

$$\hat{y}(x) = \phi(x)^T \mathbf{w}. \quad (6)$$

The weights are obtained using ridge regression:

$$\mathbf{w} = (\Phi^T \Phi + \alpha I)^{-1} \Phi^T \mathbf{y}, \quad (7)$$

where Φ is the design matrix and α is the regularization parameter. This approach follows the general idea of quantum-enhanced feature spaces, in which nonlinear mappings are produced by quantum state preparation and measurement [4, 3].

3.3. Variational Quantum Parameterization

For the synthetic geophysical inversion, a variational quantum circuit is used as a parameter generator. The circuit consists of multiple qubits, parameterized rotation gates, and entangling layers. Pauli-Z expectation values are decoded into physical parameters:

$$p_j = p_j^{\min} + \frac{\langle Z_j \rangle + 1}{2} (p_j^{\max} - p_j^{\min}). \quad (8)$$

Since $\langle Z_j \rangle$ lies between -1 and 1 , the decoded parameter is automatically restricted to a prescribed physical interval. This is useful in geophysical inversion because the search space must exclude unrealistic density contrasts, fault positions, or depths. The variational circuit therefore acts as a constrained nonlinear parameterization of the inverse problem [2].

3.4. Optimization and Uncertainty

The inversion objective is multimodal, so a single starting model may converge to a local minimum. To reduce this risk, Latin Hypercube Sampling is used to generate diverse initial models. The best solutions are then compared using the combined gravity-radiometric loss. After the optimum is obtained, Monte Carlo perturbation is used to estimate uncertainty. In each Monte Carlo realization, the observed data are perturbed within their noise level and the inversion is repeated or locally re-evaluated. The spread of recovered parameters provides an estimate of solution stability.

4. Stage 0: Quantum Regression

4.1. Purpose of the Regression Stage

Stage 0 was designed as a controlled preliminary test of the quantum regression framework. Before applying quantum methods to a geophysical inverse problem, it is necessary to verify that the implementation can generate quantum features, fit parameters, and

evaluate performance on unseen data. Two functions were considered: a linear function and a cubic function. In both cases, 100 data points were generated and split into 80% training data and 20% testing data.

4.2. Quantum Linear Regression

The linear target function was

$$y = 1.85x - 0.35 + \eta, \quad (9)$$

where η is Gaussian noise. The quantum feature map used two qubits. The scalar input was encoded through rotation gates, followed by an entangling operation. The measured expectation values were combined with the original input to form a regression feature vector. The final readout weights were solved classically.

Table 1: Stage 0 quantum linear regression metrics.

Split	RMSE	MAE	R^2
Training	0.0533	0.0438	0.9974
Testing	0.0656	0.0575	0.9960

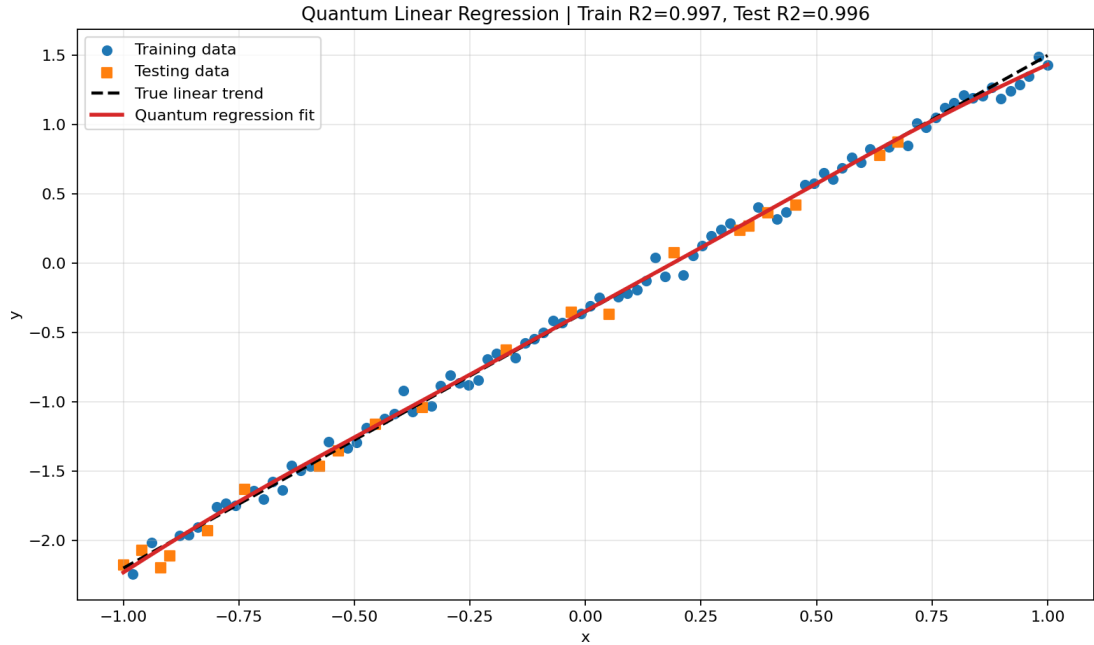


Figure 1: Quantum linear regression on generated data. The fitted model follows both training and testing points closely, indicating that the quantum feature representation is stable for the linear case.

4.3. Quantum Cubic Regression

The cubic target function was

$$y = 0.65x^3 - 0.45x^2 + 0.85x + 0.25 + \eta. \quad (10)$$

The cubic case is more demanding because the model must represent curvature and changing slope. A four-qubit feature map was used, with input-dependent rotations based on x , x^2 , x^3 , and $\sin(\pi x)$. The feature vector included polynomial terms, single-qubit expectation values, and pairwise expectation products.

Table 2: Stage 0 quantum cubic regression metrics.

Split	RMSE	MAE	R^2
Training	0.0848	0.0662	0.9965
Testing	0.1015	0.0819	0.9938

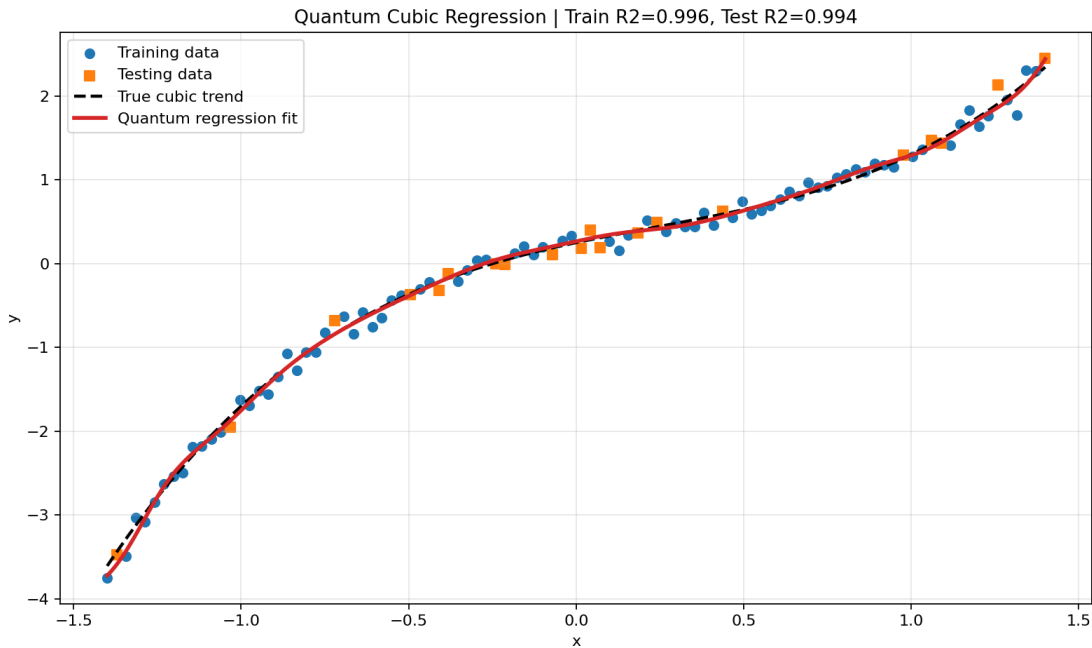


Figure 2: Quantum cubic regression result. The four-qubit feature map captures the nonlinear cubic trend and retains strong testing performance.

4.4. Interpretation of Stage 0

The regression experiments show that the feature-map implementation is able to reproduce both simple and nonlinear functions. The testing R^2 values remain close to the training values, indicating that the fitted models are not simply memorizing the training points. This result supports the use of quantum feature maps as nonlinear basis functions in later inverse modeling stages.

5. Stage 1: Synthetic Model Training and Forward Modeling

5.1. Synthetic Geological Model

Stage 1 uses a 10 km long and 2.5 km deep synthetic geological section. The model contains four units:

- **Host rock:** the reference background material.
- **Dense body:** a positive density-contrast body placed mainly on the left side of the profile.
- **Faulted basin:** a lower-density basin unit whose boundary is controlled by an inclined fault.
- **Soil layer:** a shallow layer that influences the radiometric response.

The synthetic target parameters were fault location $x_f = 4500$ m, dense-body density contrast $\rho_d = 0.200$ g/cc, basin-density contrast $\rho_b = 0.150$ g/cc, and dense-body top depth $z_d = -1000$ m. These values were chosen to produce a distinct gravity anomaly and a clear radiometric contact while remaining within plausible geophysical ranges.

5.2. Forward Modeling Procedure

The gravity response was calculated from the discretized density model. Each cell contributes to the anomaly according to its density contrast and geometric position relative to the observation points. The total anomaly is the sum of all cell contributions. Radiometric responses were generated by assigning K, U, and Th values to the geological units and smoothing the transitions to represent finite survey resolution.

Noise was added to the synthetic observations in order to make the inversion more realistic. This is important because a noise-free synthetic inversion may appear successful even when the method is too sensitive for field use. The synthetic experiment therefore tests whether the method can recover the main parameters under controlled but imperfect conditions.

5.3. Inversion Parameters and Quantum Decoding

The inversion solves for four parameters:

1. fault position x_f ,
2. dense-body density contrast ρ_d ,
3. basin-density contrast ρ_b ,
4. dense-body top depth z_d .

These parameters are decoded from expectation values of an 8-qubit, 4-layer variational circuit. The bounded decoding step ensures that each recovered value remains physically meaningful. Latin Hypercube Sampling provides several starting points, and Monte Carlo perturbation estimates uncertainty around the best solution.

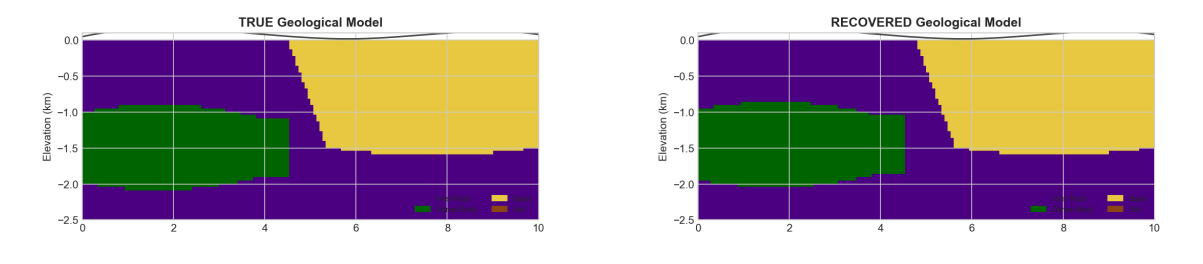


Figure 3: Stage 1 synthetic true and recovered geological sections. The recovered model reproduces the dense body, basin boundary, and main fault geometry.

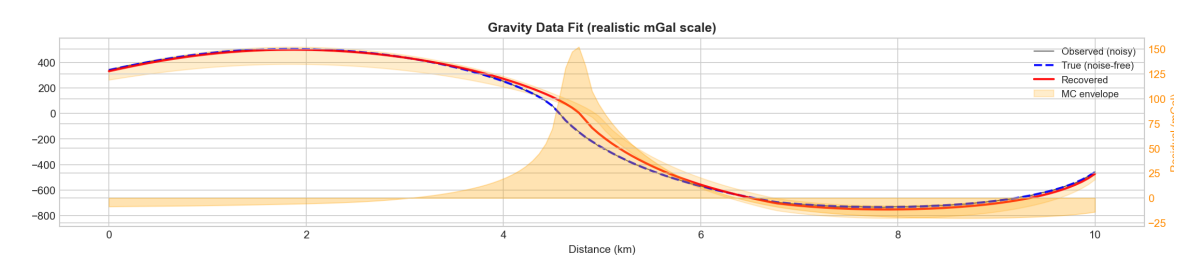


Figure 4: Stage 1 synthetic gravity response. The recovered response follows the true and noisy anomaly over the 10 km profile.

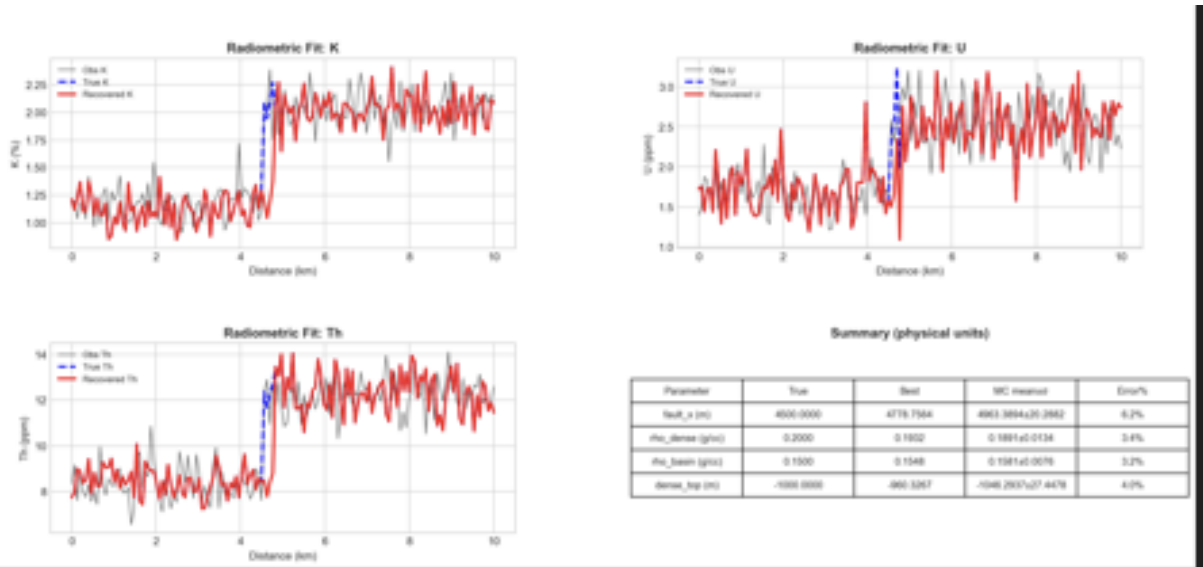


Figure 5: Stage 1 synthetic K, U, and Th radiometric fits. The radiometric curves constrain the shallow contact and near-surface contrast.

Table 3: Stage 1 synthetic parameter recovery.

Parameter	True value	Best recovered value	Relative error
Fault position	4500 m	4778.8 m	6.2%
Dense-body contrast	0.200 g/cc	0.193 g/cc	3.4%
Basin contrast	0.150 g/cc	0.155 g/cc	3.2%
Dense top depth	-1000 m	-960 m	4.0%

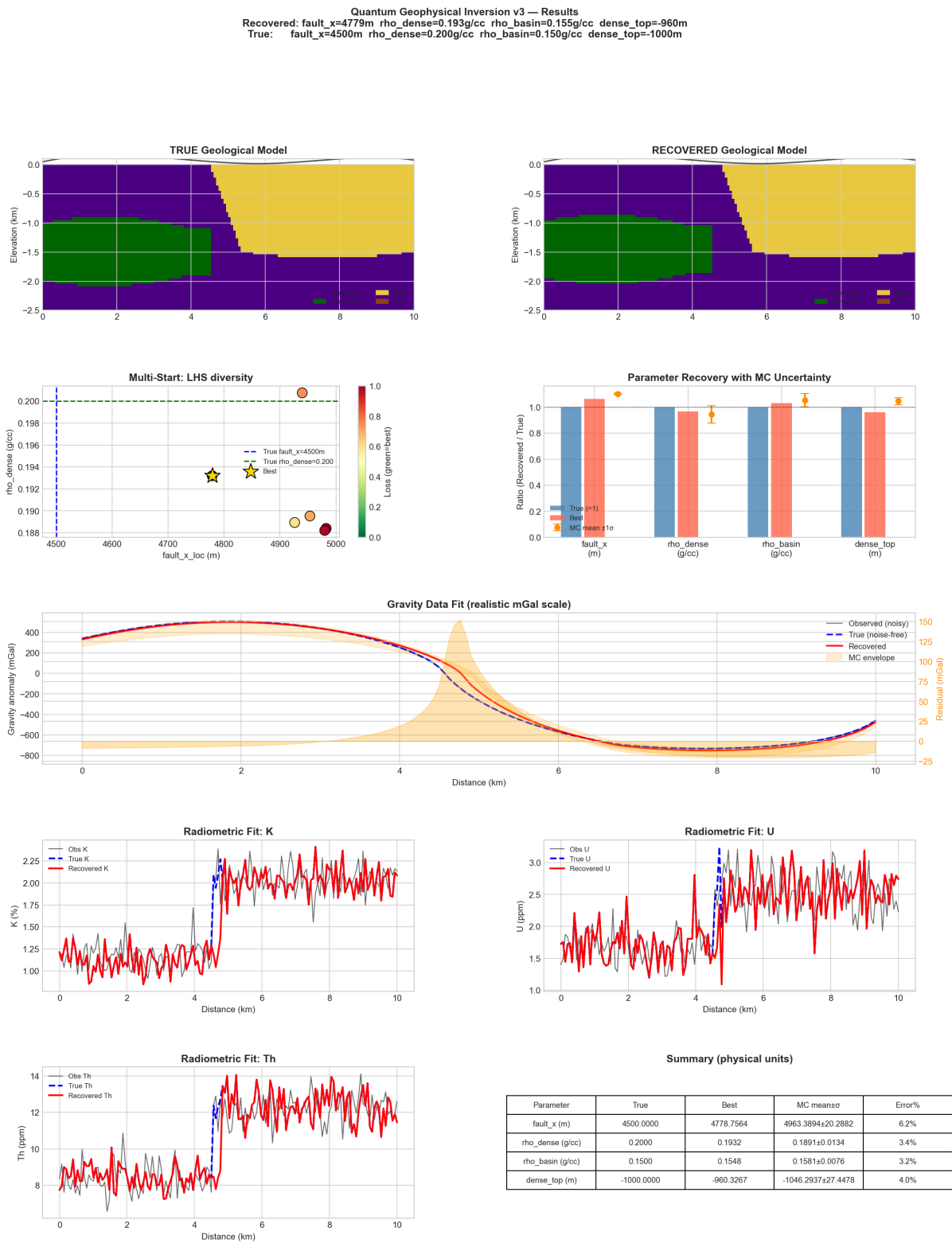


Figure 6: Complete Stage 1 synthetic inversion result page, including model recovery, multi-start behavior, parameter uncertainty, and data fits.

5.4. Interpretation of Stage 1

The synthetic experiment demonstrates that the inversion can recover the main geological structure when the forward model and target data are generated from the same physical assumptions. The fault position is the least accurate of the four primary parameters, with an error of approximately 6.2%. This is expected because the gravity anomaly is affected by both the fault boundary and the dense body, so small trade-offs between geometry and density can produce similar responses. The density contrasts and dense-body top depth are recovered more accurately, with errors below 5%. This stage therefore provides a controlled validation of the inversion framework before applying it to field data.

6. Stage 2: Real-Data Testing on a 10 km Window

6.1. Reason for Calibration

The first direct application of the synthetic inversion to the real profile was not satisfactory. The reason is that the real data do not share the exact amplitude scale, polarity, background trend, or radiometric behavior assumed in the synthetic model. A field Bouguer anomaly contains regional components that are unrelated to the local 10 km model. In addition, radiometric data are influenced by soil cover, weathering, topographic exposure, and local enrichment. Therefore, the real-data inversion required a calibrated workflow rather than a direct transfer of the synthetic model.

The calibrated workflow includes:

- robust polynomial removal of the regional Bouguer trend;
- automatic screening of possible 10 km windows;
- optional reversal of the profile direction;
- scale-offset calibration for gravity, K, U, and Th;
- a radiometric model capable of representing localized shallow enrichment;
- Monte Carlo uncertainty estimation for the recovered parameters.

6.2. Preprocessing of the Field Profile

The complete field profile is approximately 30 km long. A robust degree-2 polynomial was fitted to the Bouguer anomaly and subtracted to obtain the residual anomaly. The 0–10 km window was selected as the best single-window inversion target. K, U, and Th data were smoothed and resampled onto the model grid. The local model coordinate system was then defined from 0 to 10 km within the selected window.

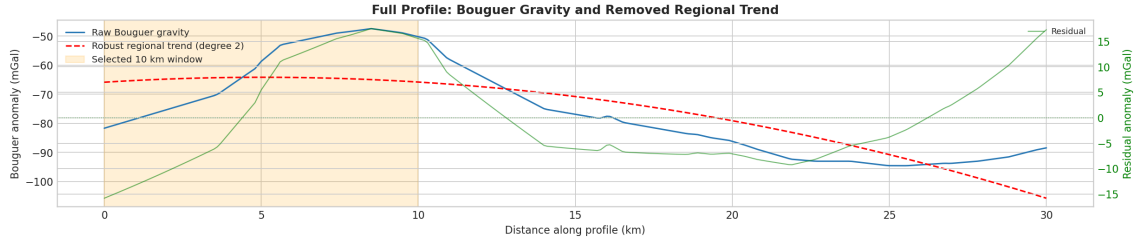


Figure 7: Full Bouguer gravity profile, robust regional trend, residual anomaly, and selected 10 km inversion window.

6.3. Recovered 10 km Geological Model

The best 10 km inversion selected the 0–10 km window without profile reversal. The recovered model places the fault near 1.50 km in local coordinates. The effective dense-body contrast is 0.2822 g/cc, the effective basin contrast is 0.0398 g/cc, and the dense-body top depth is approximately -2253 m. These values should be interpreted as calibrated effective model parameters rather than direct laboratory measurements. The reason is that the inversion includes survey calibration and simplified geometry, so the recovered contrasts represent the density differences required by the forward model to fit the observed residual anomaly.

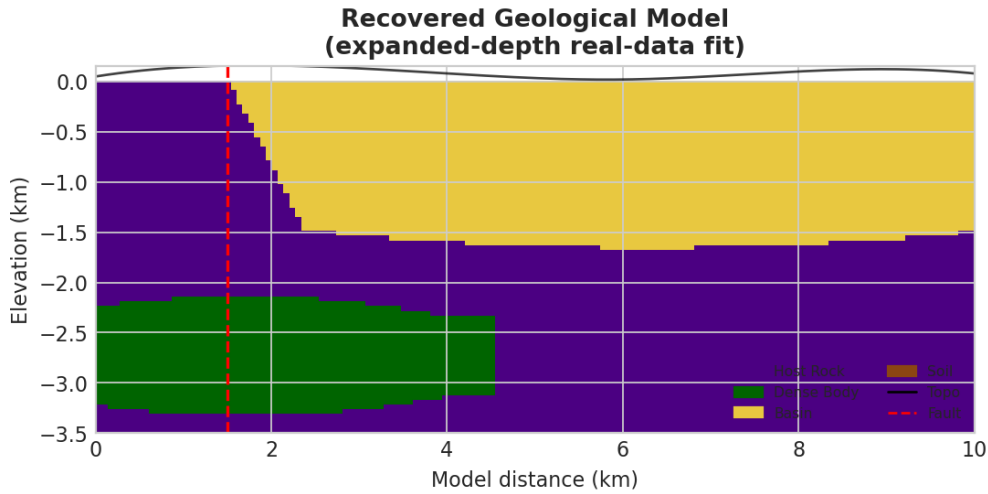


Figure 8: Recovered geological section for the selected 10 km real-data inversion.

6.4. Gravity and Radiometric Fit

The calibrated gravity response gives $R^2 = 0.989$ and $\text{RMSE} = 1.267$ mGal. This indicates that the recovered model reproduces the main residual gravity shape in the selected window. The radiometric fits are also strong, with $R^2 = 0.951$ for K, 0.929 for U, and 0.891 for Th. The Th fit is slightly weaker because the observed Th curve contains a localized response that is difficult to reproduce with a smooth model.

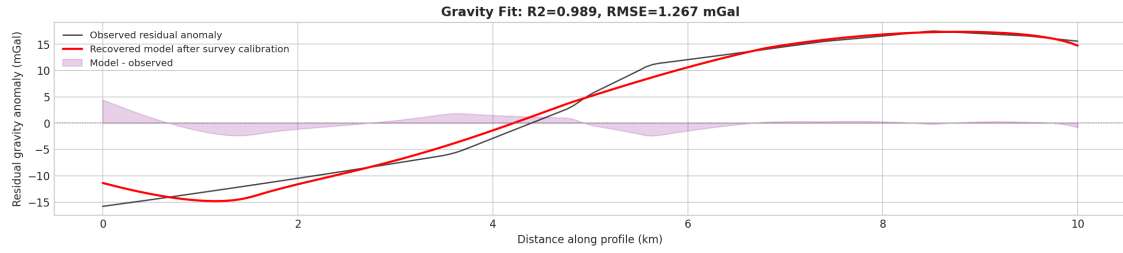


Figure 9: Stage 2 gravity fit for the selected 10 km real-data window.

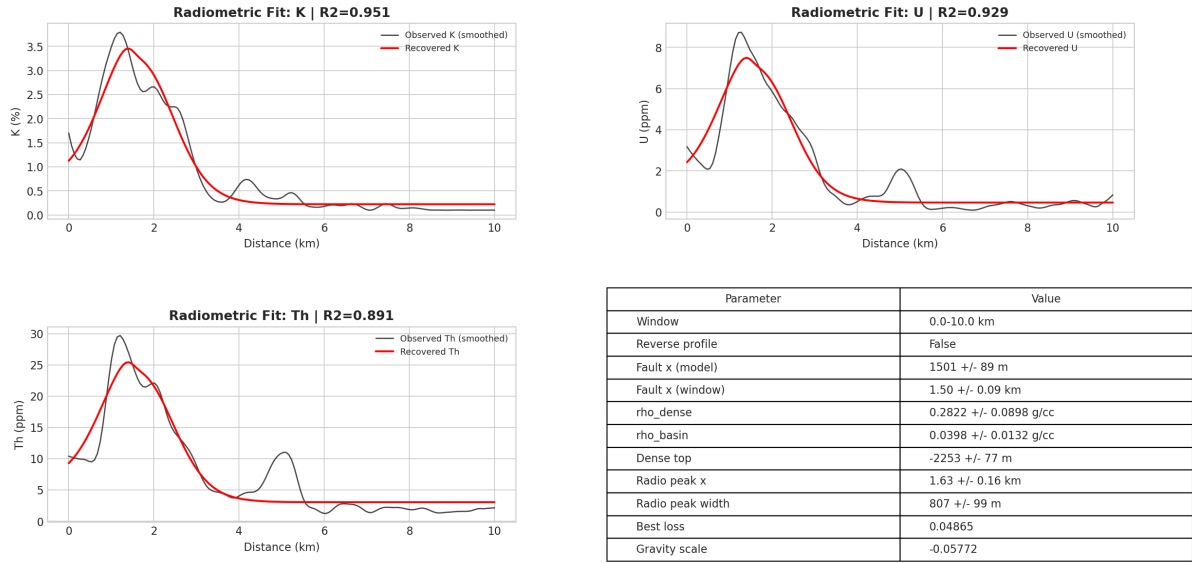


Figure 10: Stage 2 K, U, and Th fits for the selected 10 km real-data inversion.

Table 4: Stage 2 real-data 10 km inversion summary.

Quantity	Value
Selected window	0–10 km
Profile reversed	False
Best loss	0.04865
Fault position	1500.8 m
Dense-body contrast	0.2822 g/cc
Basin contrast	0.0398 g/cc
Dense top depth	-2253 m
Gravity R^2 / RMSE	0.989 / 1.267 mGal
K R^2 / RMSE	0.951 / 0.238
U R^2 / RMSE	0.929 / 0.633
Th R^2 / RMSE	0.891 / 2.541

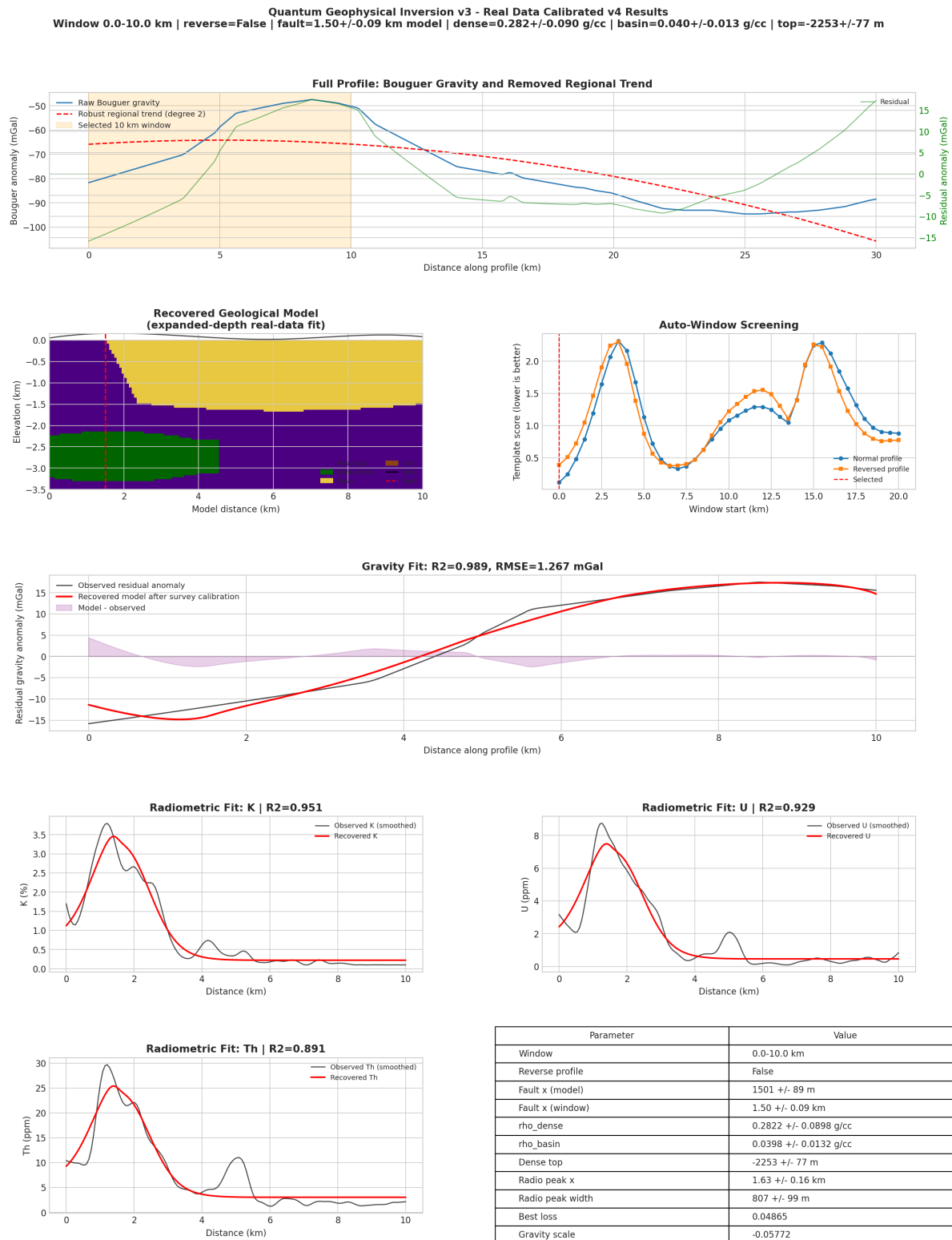


Figure 11: Complete Stage 2 result page for the selected 10 km real-data inversion.

6.5. Interpretation of Stage 2

The calibrated 10 km inversion is the most internally consistent real-data result because it uses one local coordinate system, one local calibration, and one selected anomaly

window. The strong gravity and radiometric statistics suggest that the model captures the dominant geological signal in the 0–10 km interval. However, because the inversion is simplified to a 2.5D geometry and uses effective contrasts, the recovered density values should be interpreted cautiously. The main reliable outputs are the anomaly geometry, the approximate fault location, the relative density contrast pattern, and the consistency between gravity and radiometric response.

7. Stage 3: Full 30 km Profile Through Three-Window Stitching

7.1. Need for Window-Based Extension

The real-data forward model was designed for a 10 km local section. Extending the same model directly to the full 30 km profile would require a larger geometry, additional bodies, stronger regularization, and a more complex treatment of the regional field. Instead, the full profile was interpreted through three independent 10 km windows: 0–10 km, 10–20 km, and 20–30 km. Each window was inverted separately, and the recovered sections were then placed in their global distance positions.

This procedure produces a stitched interpretation rather than a single globally coupled inversion. The distinction is important. In a global inversion, all windows would be solved simultaneously with continuity constraints across window boundaries. In the present approach, each window has its own calibration and local optimum. Therefore, the stitched section is useful for profile-scale visualization, but discontinuities near 10 km and 20 km should not be overinterpreted unless supported by independent geological evidence.

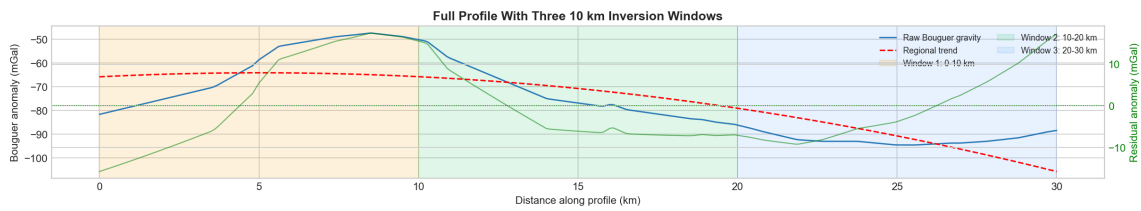


Figure 12: Full 30 km Bouguer profile with three inversion windows and the regional trend.

7.2. Stitched Geological Interpretation

The stitched geological section contains three local models arranged along the full profile. The interpreted fault locations are approximately 3.25 km, 11.54 km, and 28.33 km in global distance coordinates. The third window is fitted in reversed orientation, indicating that the local anomaly shape is better represented when the model coordinate direction is reversed. This reversal is a calibration choice for matching profile polarity and does not by itself imply a reversal of geological time or structural sequence.

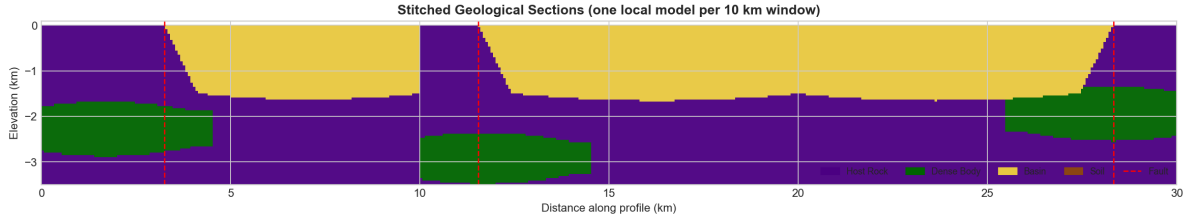
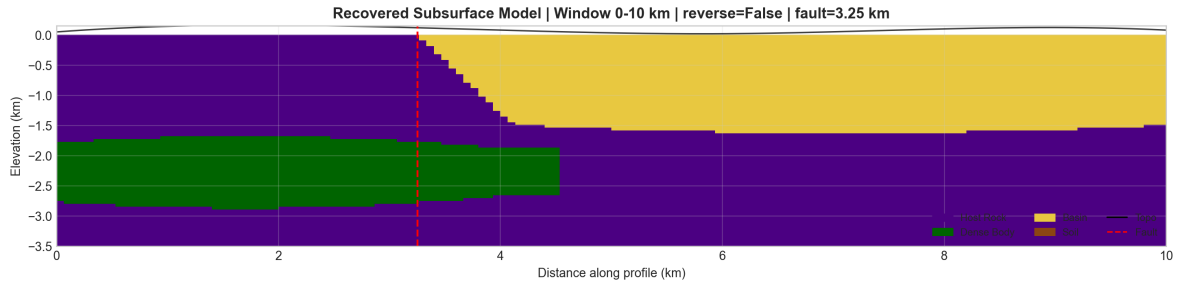
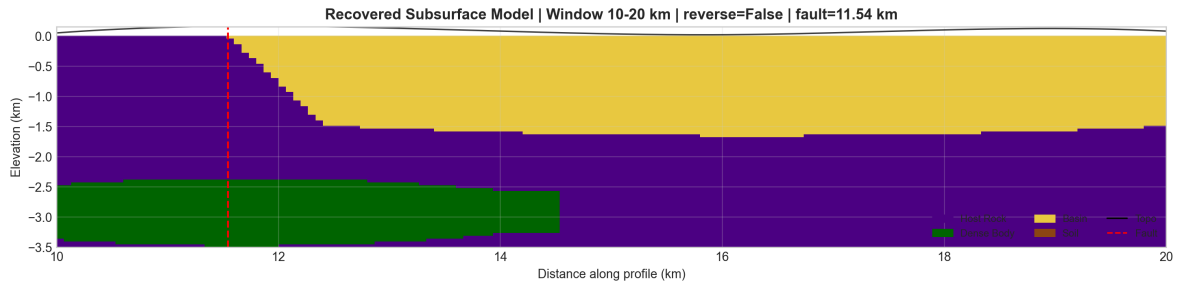


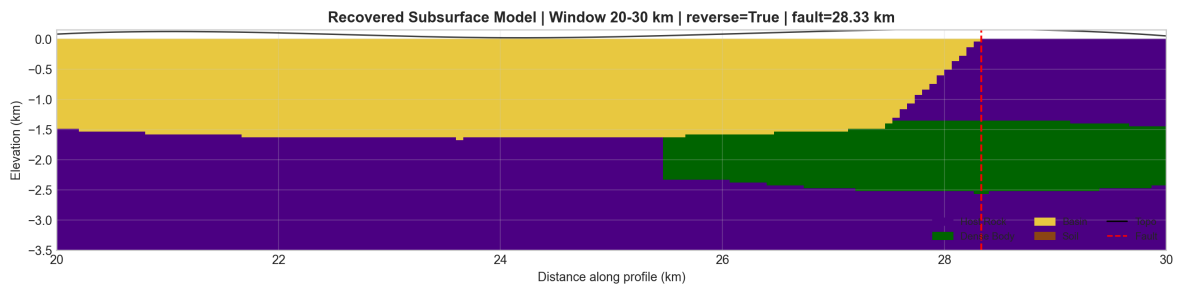
Figure 13: Stage 3 stitched geological section across the 30 km profile. Each 10 km interval is independently inverted and then placed in global distance coordinates.



(a) Window 1: 0–10 km.



(b) Window 2: 10–20 km.



(c) Window 3: 20–30 km.

Figure 14: Individual recovered geological sections for the three Stage 3 windows.

7.3. Unified Data Fit

After placing the three local responses in global coordinates, the stitched gravity response gives $R^2 = 0.965$ and $\text{RMSE} = 1.756$ mGal. The stitched radiometric fits give $R^2 = 0.883$ for K, 0.876 for U, and 0.866 for Th. These values are lower than the best single-window

result but remain strong for a 30 km profile containing multiple local anomaly regimes.

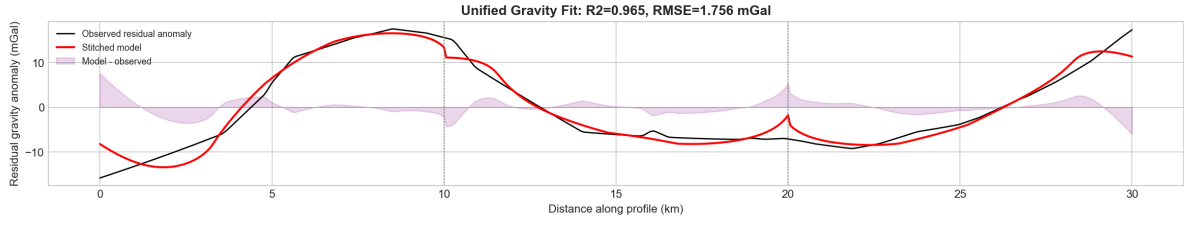


Figure 15: Unified Stage 3 gravity fit across the complete 30 km profile.

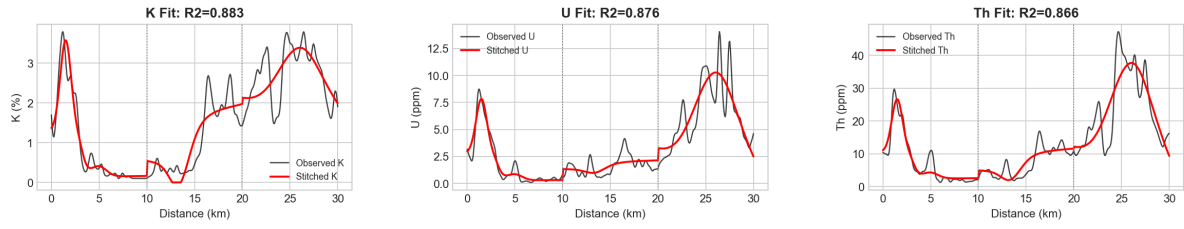


Figure 16: Unified Stage 3 radiometric fits for K, U, and Th.

Table 5: Stage 3 full-profile stitched inversion metrics.

Metric	Value
Unified gravity R^2	0.9650
Unified gravity RMSE	1.7556 mGal
Unified K R^2	0.8832
Unified U R^2	0.8764
Unified Th R^2	0.8663

Table 6: Stage 3 recovered window parameters.

Window	Loss	Reversed	Fault	ρ_d	ρ_b	Dense top
0–10 km	0.0872	False	3.25 km	0.103	0.022	-1784 m
10–20 km	0.2616	False	11.54 km	0.052	0.041	-2472 m
20–30 km	0.2202	True	28.33 km	0.315	0.158	-1450 m

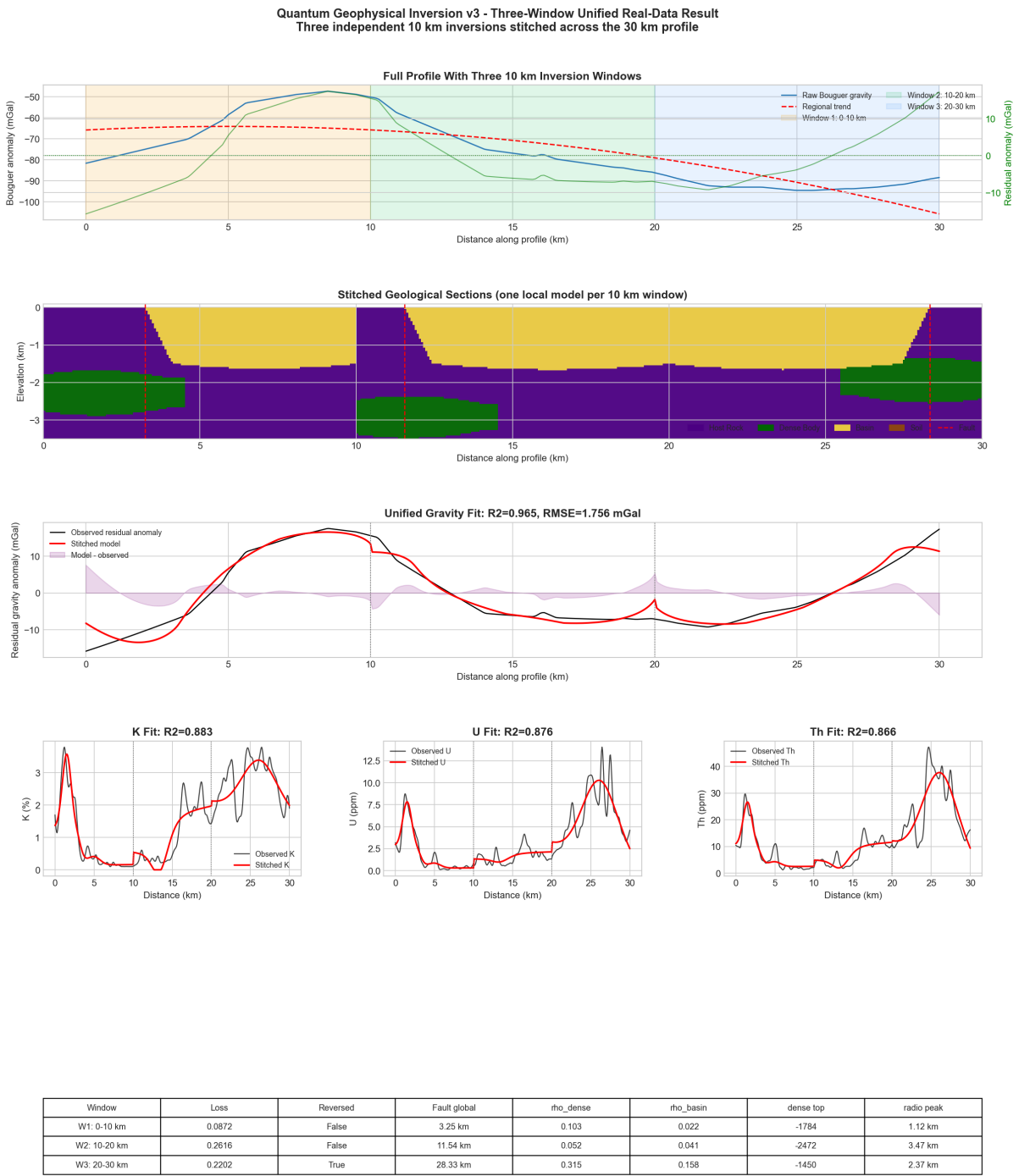


Figure 17: Complete Stage 3 full-profile stitched inversion result page.

Window 0-10 km Individual Inversion Profile

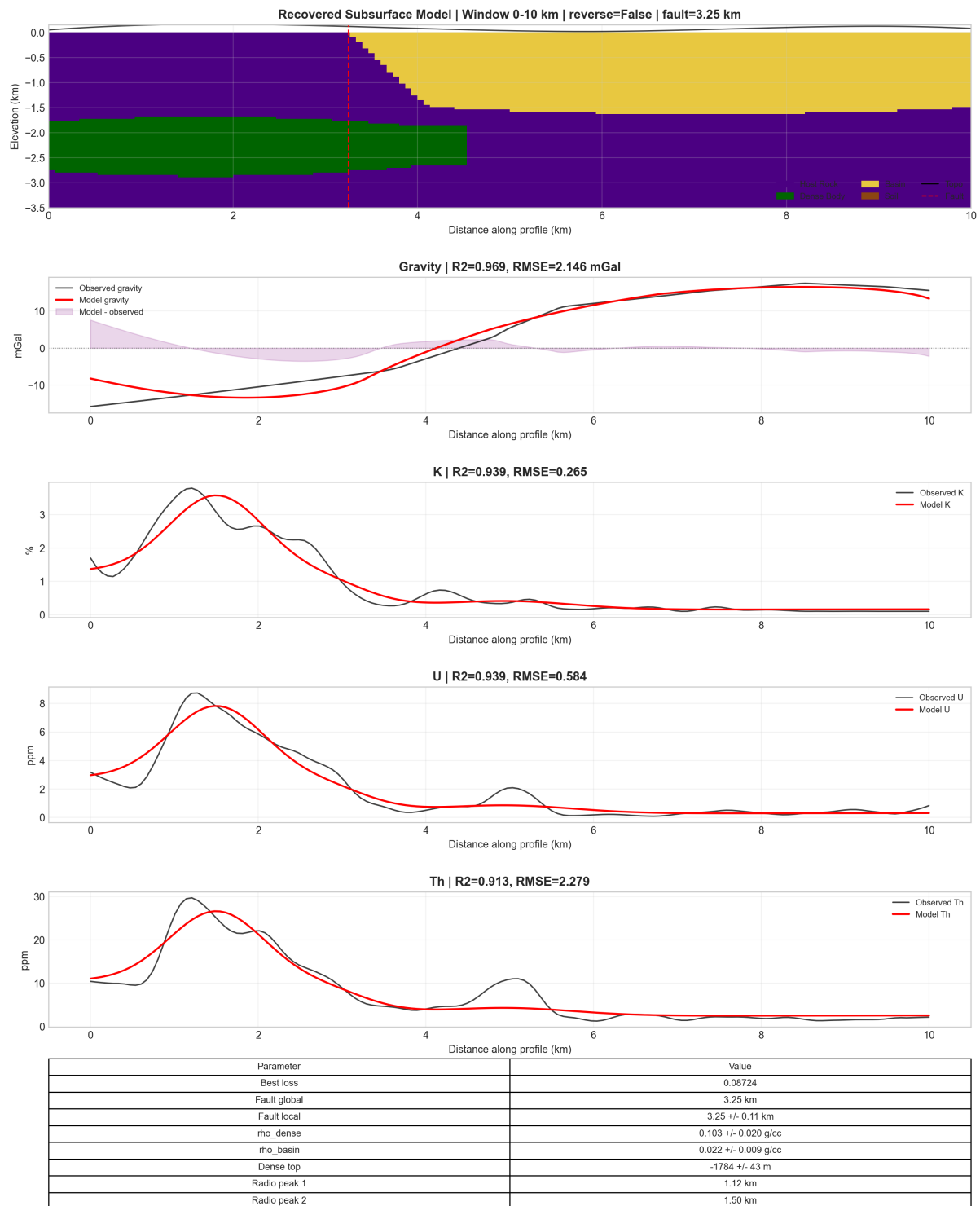


Figure 18: Complete individual inversion output for Window 1, 0–10 km.

Window 10-20 km Individual Inversion Profile



Figure 19: Complete individual inversion output for Window 2, 10–20 km.

Window 20-30 km Individual Inversion Profile

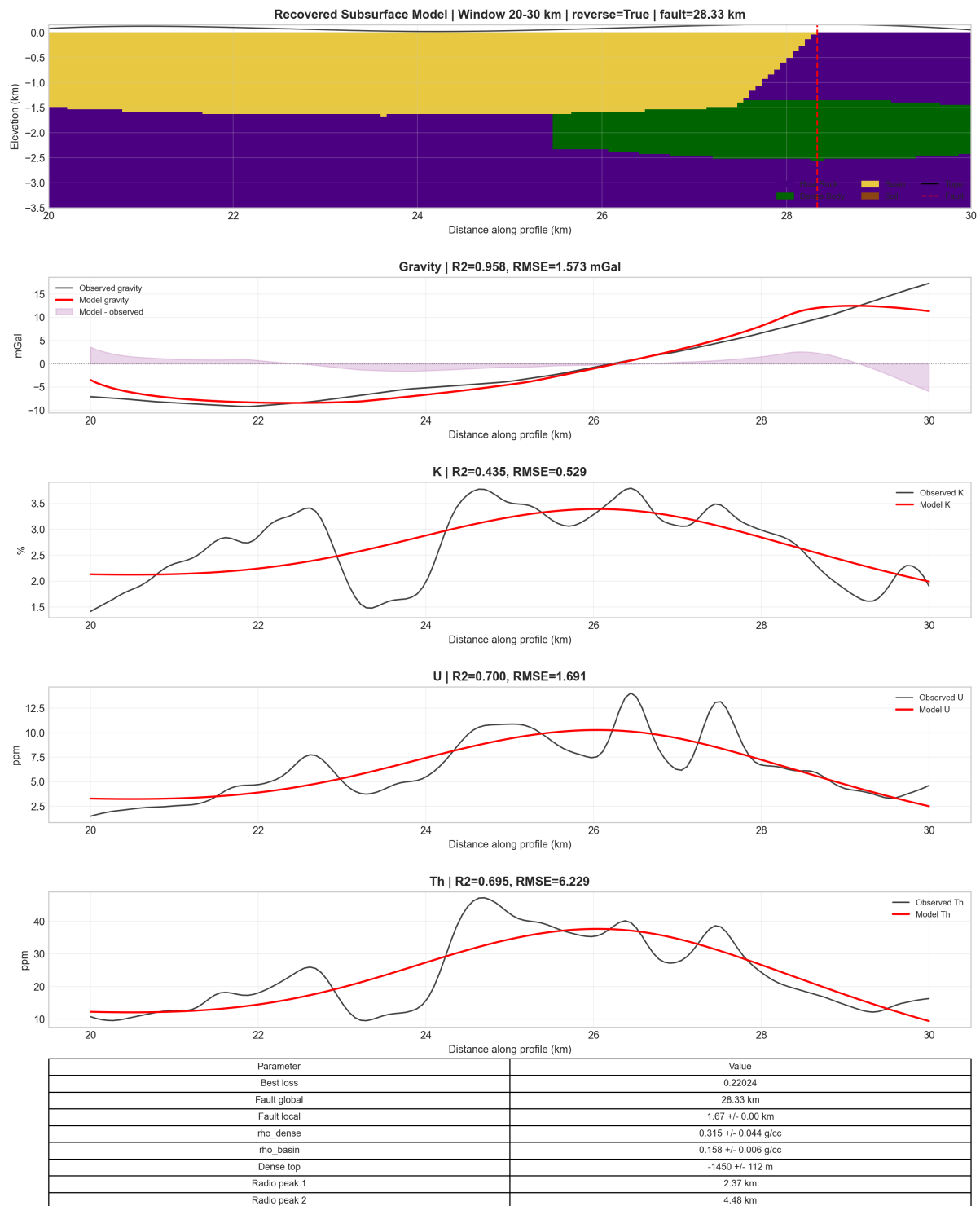


Figure 20: Complete individual inversion output for Window 3, 20–30 km.

7.4. Interpretation of Stage 3

The stitched full-profile result provides a broader geological picture than the single-window result. The first window corresponds to the strongest and most stable fit. The second and third windows contain more complicated radiometric behavior and have higher losses, which suggests that a single local body-and-basin model is less complete in these intervals. Nevertheless, the unified fit remains quantitatively strong and the stitched model provides a coherent first-order interpretation of the 30 km profile.

8. Computational Implementation

8.1. Project Structure

The computational work is organized into four stage folders:

```
Quantum Inversion/
  00_Quantum_Regression/
  01_Model_Training_and_Forward_Modeling/
  02_Real_Data_Testing_10km_True_Inversion/
  03_Real_Data_Testing_Full_Profile_Three_Window_Stitched/
```

Each stage contains the relevant source code, input data or generated data, run scripts, and result files. The stage organization is useful because the workflow progresses from a simple quantum regression test to synthetic inversion, then to calibrated 10 km real-data inversion, and finally to full-profile stitched interpretation.

8.2. Execution on Param Shakti

The geophysical inversion stages are suitable for execution on the Param Shakti high-performance computing facility using Slurm scripts. The computationally expensive parts are the multi-start optimization and Monte Carlo uncertainty estimation, both of which require repeated forward-model evaluations. The scripts expose parameters such as maximum iterations, population size, Monte Carlo sample count, and radiometric weighting. This allows the same code to be run quickly for testing or more extensively for final results.

The regression stage is lightweight and depends only on standard numerical libraries. The geophysical quantum inversion stage uses PennyLane for the variational quantum circuit. If an optimized simulator backend is unavailable, the code can use a default qubit simulator. This design makes the workflow portable across local computers and HPC environments.

9. Discussion

9.1. Geophysical Significance

The results show that calibration is essential when transferring a synthetic inversion framework to field data. The synthetic model validates the forward and inverse formulation under known conditions, but the field profile contains regional gravity, survey-specific scaling, noise, and shallow radiometric effects. Once these factors are included, the inversion produces much more satisfactory results.

The 10 km real-data inversion is the strongest real-data result because it focuses on one coherent anomaly window. The full-profile stitched result is useful for visualizing the complete line, but it should be interpreted as a composite of local inversions. The recovered fault positions and dense-body geometries are therefore best treated as first-order structural indicators rather than final geological boundaries.

9.2. Quantum Significance

The quantum part of the project has two roles. First, the regression stage demonstrates that quantum feature maps can represent both linear and nonlinear functions with strong testing performance. Second, the synthetic geophysical stage demonstrates that a variational quantum circuit can act as a bounded parameter generator for inversion. In this formulation, quantum expectation values are mapped to physical model parameters, and the classical optimizer adjusts the circuit weights to reduce geophysical misfit.

At the present scale, the computations are performed using quantum simulation on classical hardware. The significance is therefore not a demonstrated quantum advantage, but the development of a working hybrid quantum-classical inversion structure. This is an important step because it shows how quantum circuits can be connected to forward modeling, objective functions, uncertainty estimation, and geophysical interpretation.

9.3. Limitations

The study has several limitations:

- the gravity model is 2.5D rather than fully 3D;
- density contrasts are calibrated effective contrasts rather than direct rock-property measurements;
- the radiometric model is a shallow response proxy rather than a full radiation-transport simulation;
- the full 30 km profile is interpreted by stitched local inversions rather than a single global inversion;

- Monte Carlo uncertainty is local and does not represent a full Bayesian posterior distribution;
- the quantum computations are simulated on classical hardware.

These limitations do not invalidate the results, but they define the correct level of interpretation. The final models should be viewed as calibrated geophysical interpretations that explain the observed profiles within a simplified but physically motivated modeling framework.

9.4. Possible Future Work

Future improvements may include a globally coupled 30 km inversion with continuity constraints, a more detailed 3D density model, use of petrophysical constraints from field or laboratory measurements, and a more rigorous Bayesian uncertainty analysis. On the quantum side, the workflow can be extended to quantum kernels, amplitude encoding, hardware-executed circuits, and hybrid optimizers designed specifically for noisy quantum devices.

10. Conclusion

This work developed and tested a four-stage framework for joint gravity and radiometric forward modeling and geophysical inversion using hybrid quantum-classical methods. Stage 0 confirmed that quantum feature-map regression can learn controlled linear and cubic relationships with high testing accuracy. Stage 1 validated the synthetic geophysical inversion by recovering the main parameters of a known faulted-basin model. Stage 2 showed that real field data require regional-trend removal, survey calibration, and an improved radiometric response model; after these modifications, the selected 10 km inversion produced a strong gravity fit and high radiometric agreement. Stage 3 extended the interpretation to the full 30 km profile through three independent 10 km inversions and produced a unified stitched geological section.

The main conclusion is that a physically calibrated, survey-aware inversion strategy is necessary for real geophysical data. Quantum feature maps and variational quantum parameterization can be integrated into this workflow in a structured way, providing a controlled hybrid framework for nonlinear inverse modeling. The resulting models are not unique geological solutions, but they provide quantitatively supported and geologically interpretable explanations of the gravity and radiometric observations.

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